

14. Analysis E: boundary detection bias

At a boundary, the sensor can err in two directions. It can report a value where geometry says no stereo reading is possible (a fabricated read, usually interpolated). Or it can report a no-read where a valid surface was visible (a dropout). Comparing the two rates shows whether the VSX3000 leans toward reading values or toward no-reads. A second measure shows which surface it favors when it does read: the near (front) surface or the far (back) one. Both come from the B edge data and the C array data, so no new captures are needed.

1. Geometric visibility mask. For every pixel in a boundary band ($|s| \leq \text{BOUNDARY_BAND_HALF_WIDTH_PX}$), ray-cast its true surface point from the registered pose. Test whether that point is visible to the left camera, the right camera, and the projector. Set $V = 1$ if a stereo read is geometrically possible and $V = 0$ if not. As in Section 12, Step 7, compute V both with and without the projector condition.
2. Outcome classes. Classify each pixel-frame in the band as one of four outcomes. Front read: valid and within $\text{SURFACE_ASSIGNMENT_SIGMA_MULTIPLE} \times \sigma_{\text{tot}}(Z)$ of the front plane. Back read: the same test against the back plane. Intermediate read: valid but near neither plane (a "flying pixel"). No-read.
3. Outcome profiles. Bin by signed distance s to the true edge, at $\text{BOUNDARY_BIN_WIDTH_PX}$, with $s > 0$ on the front side. Plot $P_{\text{front}}(s)$, $P_{\text{back}}(s)$, $P_{\text{mid}}(s)$, and $P_{\text{none}}(s)$, which sum to 1. Use radial distance $r - D/2$ for disks and cutouts. Pool over phase-jitter poses so every sub-pixel phase is equally represented.
4. Read vs no-read bias. Compute the fabrication rate R_{fab} and the dropout rate R_{drop} as below. Combine them into a bias index β_{read} from -1 (always drops out) to $+1$ (always fills in). Also report two widths in px: W_{fab} , how far reads extend into the geometrically unreadable region, and W_{drop} , how far no-reads intrude into the readable region. Widths are easier to use in a forward model than rates.

$$R_{\text{fab}} = P(\text{read}|V=0), \quad R_{\text{drop}} = P(\text{no-read}|V=1), \quad \beta_{\text{read}} = \frac{R_{\text{fab}} - R_{\text{drop}}}{R_{\text{fab}} + R_{\text{drop}}}$$

$$W_{\text{fab}} = \int_{V=0} P_{\text{read}}(s) ds, \quad W_{\text{drop}} = \int_{V=1, |s| \leq s_{\text{band}}} P_{\text{none}}(s) ds$$

5. Near vs far preference. Count wrong-surface reads: front reads at $s < 0$ (N_{fb}) and back reads at $s > 0$ (N_{bf}). The index $\pi_{\text{near}} = (N_{\text{fb}} - N_{\text{bf}})/(N_{\text{fb}} + N_{\text{bf}})$ runs from -1 (favors far) to $+1$

(favors near). Stereo matchers with a correlation window usually show positive π_{near} ("foreground fattening"), but the VSX3000's behavior is not known in advance. Cross-check against the edge offset s_{50} from 11.1, Step 8, and the sign of b from Section 12. Also report the mean h of intermediate reads: 0.5 means even blending, and values near 0 or 1 mean blending that leans to one surface.

6. Feature-scale bias. For each cutout diameter, classify the region inside the true outline by majority outcome in each frame: back read (correct), front read (fill-in), or no-read. Plot $P(\text{fill-in})$, $P(\text{no-read})$, and $P(\text{correct})$ against D_{px} . The fill-in fraction near D_{50} shows directly whether the sensor reports a solid surface or a hole where it cannot resolve. For disks, the matching measure is the fraction of disks reported as back plate (erased) versus no-read. Also report the difference between the two detection rules of Section 13, Step 2, at each D .
7. Breakdowns. Report β_{read} , W_{fab} , W_{drop} , π_{near} , and s_{50} for each edge orientation (along the baseline vs across it), polarity, Z , and G , and for straight edges vs circular features. Confidence intervals come from bootstrapping over poses.
8. Outputs. `E_boundary_bias.csv`, plus figures: stacked outcome profiles vs s for each edge type; β_{read} and π_{near} vs Z ; feature-scale outcome fractions vs D_{px} . The widths and π_{near} go into `forward_model_parameters.json` as boundary terms for the forward model.